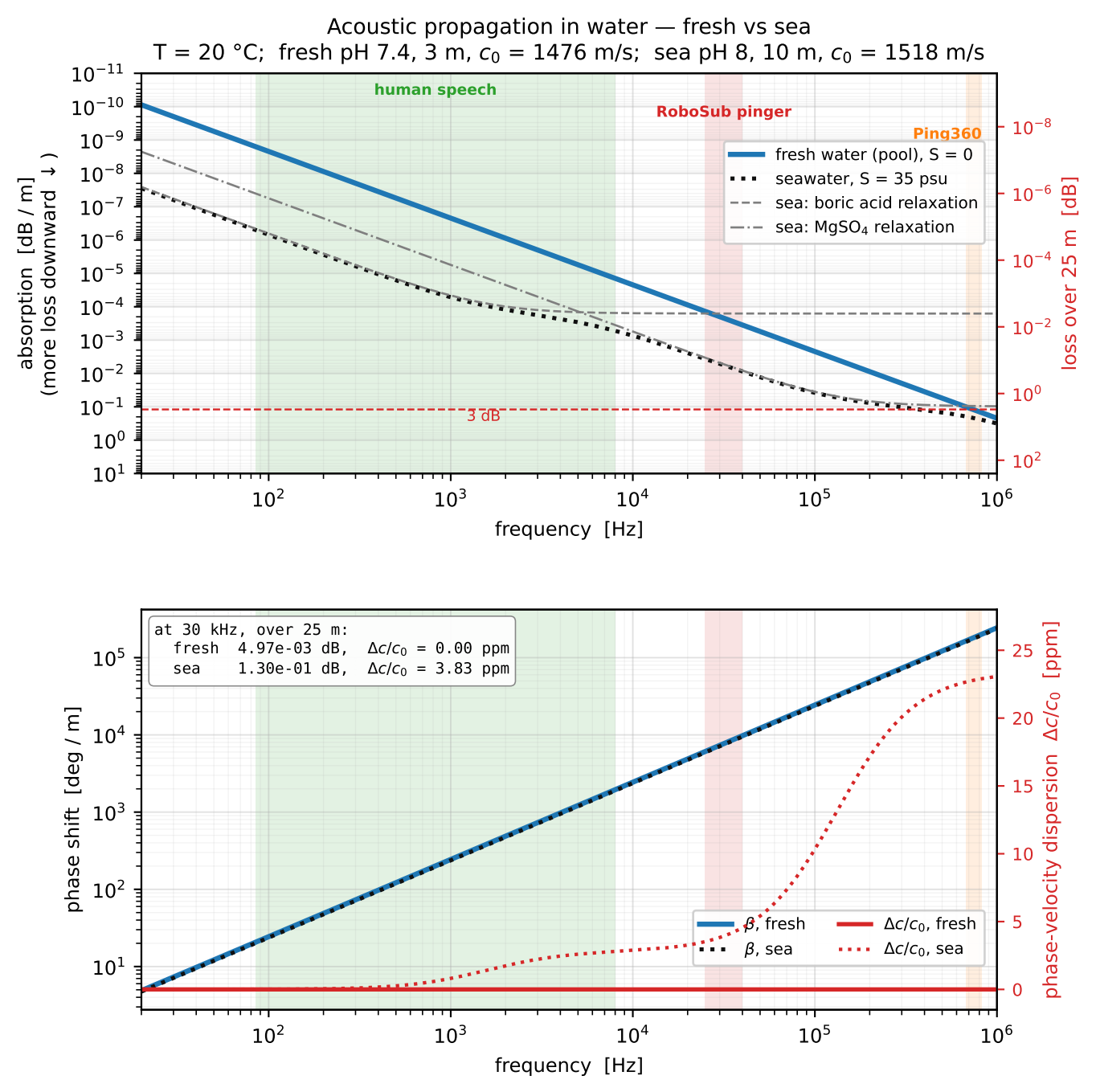
Acoustic propagation in water — fresh vs sea
T = 20 °C; fresh pH 7.4, 3 m, c_0 = 1476 m/s; sea pH 8, 10 m, c_0 = 1518 m/s
human speech
RoboSub pinger
Ping360
fresh water (pool), S = 0
seawater, S = 35 psu
sea: boric acid relaxation
sea: $MgSO_4$ relaxation
3 dB
absorption [dB / m]
(more loss downward ↓)
loss over 25 m [dB]
frequency [Hz]
at 30 kHz, over 25 m:
fresh 4.97e-03 dB, $\Delta c/c_0$ = 0.00 ppm
sea 1.30e-01 dB, $\Delta c/c_0$ = 3.83 ppm
phase shift [deg / m]
β, fresh
β, sea
$\Delta c/c_0$, fresh
$\Delta c/c_0$, sea
phase-velocity dispersion $\Delta c/c_0$ [ppm]
frequency [Hz]